# Transistors, relays & MOSFETs

ENGINEERING WITH MR KELLY

National 5 · Electronics & control · Worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ describe how a transistor works as a switch (about 0.7 V base–emitter)
- ☐ explain the purpose of a relay and a protection diode
- ☐ choose a transistor to switch a high-current output such as a lamp or motor

## I DO — Worked example

Read how a full-mark answer is built

*A microcontroller output pin cannot supply enough current to light a 12 V lamp.*

**1 Name the component that should be added to switch on the lamp.** (1 mark)

**Points that earn marks** from the marking instructions: one ✓ per point

✓ Transistor (or MOSFET)

**Why answers lose marks**

✗ Resistor, relay on its own

## WE DO — Guided practice

Use the starters and word bank to help you

*A transistor drives a relay coil. A diode is connected across the coil.*

**2 Explain the purpose of the diode.** (2 marks)

**Sentence starters**

One reason is … because …

Another reason is … so …

**Word bank**

protecting · conducts · voltage · damage · safely · produces · switches · transistor

Tip: 2 marks = 2 separate, developed points.

## YOU DO — On your own

No notes. Check your answers afterwards.

**3 Quick check: tick one box for each.**

**a)** Which component lets a 5 V circuit switch a 230 V heater?

☐ LDR ☐ Relay ☐ Diode ☐ Buzzer

**b)** Why is a diode placed across a relay coil?

☐ To light up ☐ To protect the transistor from reverse voltage ☐ To increase current ☐ To switch the relay

**c)** An NPN transistor switches on at a base voltage of about…

☐ 0.07 V ☐ 0.7 V ☐ 7 V ☐ 70 V

**4 Key terms: write the letter of the matching description next to each term.**

| | | |
|---|---|---|
| 1. **Transistor switch-on voltage** | ___ | **A** Lets a low-current circuit switch a separate high-current circuit. |
| 2. **Base resistor** | ___ | **B** Limits base current to protect the transistor. |
| 3. **Relay** | ___ | **C** About 0.7 V at the base. |

*A relay is used between a transistor and a mains lamp.*

**5 Explain why a relay is used.** (2 marks)

**6 Explain what is meant by ‘Protection diode’.** (2 marks)

**Check and reflect**

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?